



Fig. 3: Connection while the mobile node is roaming; registration request and reply for home agent, foreign agent and mobile node

between two different mobile nodes.

However, this will work only when the mobile node is available to register its mobility with the foreign agent in the foreign network. It has to register when it enters the foreign network as well as when it leaves the foreign network. This is done by a process called 'getting recognition' (discovery) and registration.

Getting recognition is a process by which a mobile node registers its location with an appropriate agent, either home agent or foreign agent. The mobile node finds a mobility agent by two methods: agent advertisement and agent solicitation message.

Agent advertisement is a process by which the foreign agent and home agent exchange their respective presence periodically using the agent advertisement message. A mobile node in a subnet then receives agent advertisements from either its home agent or foreign agent. This is one way for the mobile node to discover its location.

In the second method, a mobile node may use the agent solicitation message for quick reply for locating the agent. In either case, this process allows a mobile node to identify its current mobility agents and then obtain one or more care-of addresses. It, by the process, also learns about any special service provided by a foreign

agent and determines whether it is connected to its home network or a foreign one.

Registration is a process by which, having received a care-of address, the mobile node has to register with the home agent. As the mobile node roams, it connects to a series of one or more foreign networks. The main purpose of this registration is to inform the home agent about the current location of the mobile node for correct forwarding of packets.

This process provides a link between the home and care-of addresses and defines the lifetime of the registration. This is done by sending binding updates from the mobile node to the home agent. For this, three values—mobile node's home static IP address, care-of address and lifetime—are used. The home agent sends a registration reply to the foreign agent. The foreign agent forwards this reply to the mobile node. This operation is shown in Fig. 3.

Encryption is used to authenticate the registration information. A key concern with registration is security. By default, the authentication algorithm uses key-based MD5 (message digest version 5) to produce a 128-bit message digest.

When the mobile node is in a foreign network and registered with



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